The Index Is Stale

A structured discussion on operating a vector index that nobody maintained

Duration: 45 to 60 minutes

Group size: 3

Output: One Plan Sheet per group plus a 75-second report out

Module context: Fine-Tuning and Retrieval-Augmented Generation. This discussion sits between the measurement segment and the lifecycle segment. You are being asked to reason about index operations before you are shown the standard answers.

READ THIS FIRST This scenario is deliberately underspecified. Deciding which missing facts are worth blocking on, and which you can proceed without, is most of the work. There is no code to write and no notebook to open. You are producing a plan and then defending it.

1. How this session runs

You will work in a group of three. Three is deliberate. Nobody gets to be a passenger, every position must be held by someone, and a group of three cannot hide a disagreement behind a majority. In your roles, you will walk through and respond to each round (as described below).

2. Your role

Each person takes one role. Roles are not decoration. Each one owns a different definition of the word bad, and the friction between those definitions is where the useful part of this discussion happens. Argue your role even when you privately agree with someone else in the group.

Role	You own	The question you keep asking
On-call engineer	The pager, and the blast radius of every command anyone proposes	What happens to live traffic while this runs, and how do I undo it?
Support operations lead	What a Cordwell agent sees on screen when the answer is wrong, and the relationship with the team that owns the chat bot	Who gets hurt between now and the fix, and how many of them?
Risk and resourcing partner	What ends up in a claim file, and what two engineers and a half-time SRE can deliver before the freeze	Which of these failures do we have to answer for, and what are we giving up to do this?

3. The scenario: Cordwell Assist

Cordwell Home and Hardware runs an internal support search tool called Cordwell Assist. It sits in front of the support knowledge base and feeds three consumers: a retrieval widget inside the agent console, a Slack bot owned by the store operations team, and a nightly report of the questions that came back with nothing useful.

The index was built six months ago by an engineer who has since moved to another team. It has not been refreshed since. It has never been evaluated since launch.

Yesterday an agent escalated a ticket because Assist confidently recommended a returns process that Cordwell stopped using in March and cited a document to support it. Someone went looking. Here is what the audit turned up.

#	Finding	What is known about it
1	Roughly 15 percent of the documents in the index no longer exist in the source CMS	About 2,760 documents. Confirmed by comparing the index against last night's export.
2	An unknown number of the remaining documents have been edited since indexing	Nobody can currently say how many. Nobody has tried to find out.
3	Nobody recorded which chunk IDs belong to which document	Record IDs are UUID version 4, generated at ingest and never written down anywhere outside the index.
4	The index serves live traffic	About 40 queries per minute during business hours. Three consumers, one of which your team does not own.
5	The embedding model has a newer version available	It is a different dimension from the one currently in production.

YOUR ASSIGNMENT Produce a plan. What do you do first, what can wait, and what do you refuse to do without more information?

4. Environment facts

Everything below is established fact for the duration of this discussion. Do not invent facts that contradict it. Do notice what is absent.

Item	Value
Index	cordwell-support-prod, Pinecone serverless, AWS, us-east-1
Metric and dimension	cosine, 768 dimensions
Namespaces	support (agent console and Slack bot), internal_ops (store associates)
Records in support	31,600 vectors, built from 18,400 source documents
Records in internal_ops	8,100 vectors, built from 4,900 source documents
Chunking at build time	Fixed 1024 tokens, no overlap, no attempt at semantic boundaries

Item	Value
Record IDs	UUID version 4, generated at ingest, not recorded anywhere else
Embedding model in production	Self-hosted sentence-transformers, 768 dimensions, one CPU worker. Measured ingest throughput at build time was about 340 chunks per minute.
Proposed replacement model	Same family, newer version, 1024 dimensions
Source of truth	Cordwell Docs CMS. Exported to S3 as markdown every night at 02:00.
Query volume	About 40 per minute during business hours, peaking near 140 per minute on Saturday mornings. Roughly 400,000 queries per month.
Backups	None configured
Deletion protection	Off
Monitoring in place	p99 query latency dashboard, 5xx error alerting
Retrieval quality monitoring	None
Labeled query set	The launch deck claims recall@5 of 0.86. Nobody can find the query set that produced that number.
Team	Two engineers and a half-time SRE
Calendar	The retail change freeze starts in six weeks and runs through January

Cordwell Home and Hardware is fictional. These figures are synthetic and internally consistent. Treat them as real for the next hour.

5. What is stored on each vector

This is one record as it exists in the index today. It is representative of all 39,700.

```
id = "9f2c1e84-7b6a-4d31-a0c5-2e5b8d41f7aa"      # UUID v4

metadata = {
    "text":         "...the chunk text itself...",
    "source_path": "troubleshooting/thermostat/th40-wifi-dropout.md",
    "chunk_index": 3,
    "ingested_at": 1769051460,   # epoch seconds: when WE indexed it
}
```

DO NOT SKIP THIS Read the schema before you start Round 1. It contains both the reason this is hard and at least one thing that makes it easier than it looks. Several of the questions ahead are unanswerable if you have not looked at it carefully.

6. Known, unknown, and what an answer would cost

What you know	What you do not know	What an answer would cost
15 percent of documents are gone from the CMS	Which specific documents, and which vectors belong to them	Low. An hour of scripting against the nightly export.
Record IDs are UUIDs	Whether any chunk can be traced back to a document at all	Minutes. Reread Section 5.
The index serves live traffic	Whether that is a capacity constraint or a correctness constraint	Minutes. Compare 40 per minute against Section 7.
Documents have been edited	How many, and whether the edits are material	Low to moderate. A sample audit gets you an estimate today.
A newer embedding model exists	Whether it is better on Cordwell's corpus	High. Requires a labeled query set you do not have.
Recall@5 was 0.86 at launch	What recall@5 is right now	High, and for the same reason.

7. Platform constraints that apply

These are current Pinecone serverless limits, confirmed against Pinecone's published documentation on 29 July 2026. Your plan must live inside them. At least one of them will change what your group's instinct was.

Operation	The limit that matters here
Delete by record ID	Up to 1,000 IDs per request. 100 delete requests per second per namespace. 5,000 records per second per namespace.
Delete by metadata filter	Supported on serverless. 5 requests per second per namespace, 500 per second per index.
Fetch by metadata filter	Up to 10,000 records per response and 4 MB per response. 5 requests per second per namespace. Paginate for more.
List record IDs by ID prefix	200 list requests per second per index.
Query	100 requests per second per namespace. Maximum top_k of 10,000. Maximum result size 4 MB.
Upsert	1,000 records or 2 MB per batch. 100 upsert requests per second per namespace. 50 MB per second per namespace.
Metadata filter operators	A single $in or $nin array accepts at most 10,000 values.
Metadata size	40 KB of filterable metadata per record. Record ID maximum 512 characters.
Index configuration	Dimension and metric are fixed at index creation and cannot be changed afterward.
Backup and restore	A backup is a static, non-queryable copy. Restoring creates a new index with the same dimension, metric, cloud, and region as the source. A backup only contains vectors that were in the index at least 15 minutes before it ran. Serverless backups are a Standard and Enterprise plan feature.

Operation	The limit that matters here
Consistency	Pinecone is eventually consistent. There is a delay between a write being accepted and it being visible to queries.

8. Round 0: your first move

Silent. Individual. Do not discuss.

Before anyone in your group speaks, write two things.

- The first thing you would do tomorrow morning. Not a strategy. One action, specific enough that a colleague could execute it without asking you a follow-up question.
- The fact you are assuming is true for that to be the right first action.

You will compare these at the start of Round 1 and again at the end of the session. Groups routinely discover that four capable engineers picked four different first moves for four different unstated reasons.

My first action

The fact I am assuming

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9. Round 1: triage

Group.

You cannot fix everything at once, and these five findings are nowhere near equally urgent.

1.1 Rank the five findings by harm per hour to a Cordwell customer, not by how hard they are to fix. Write the ranking down. Then name the single fact that would reorder your list and say who in the company already has that fact.

1.2 One of these findings produces a confident wrong answer with a citation attached. One produces a mildly degraded answer. One produces no user-visible symptoms at all today. Assign every finding to one of those three categories. Report the assignment your group argued about, not the ones that were obvious.

1.3 Take a document that was deleted from the CMS but is still sitting in the index. Construct one case where this is the single most dangerous item on the whole list, and one case where it is completely harmless. What distinguishes them? Your answer should be a property, not an example.

1.5 Pick the cheapest measurement that converts at least two unknowns into knowns. State the sample size, the wall-clock time to run it, and what precision that sample buys you. If your group cannot state a number, you have picked a strategy rather than a measurement. Go again.

10. Round 2: the Plan Sheet

Group.

Action	Owner role	Why now, or why not yet	Risk and how you undo it
Next 4 hours			
This week			
This quarter			

What we refuse to do until we know more

What we refuse to do	The fact that would unblock it	Who has that fact

11. Round 3: report out

#	State this	What makes it count
1	Your first action	A command or a concrete task, not a strategy. Someone should be able to execute it on Monday without asking a question.
2	The one thing you refuse to do, and the single fact that would change your mind	Name the fact and name who has it. "More information" is not a fact.
3	Where your group genuinely disagreed, and whether you resolved it	Not optional. "We agreed on everything" is not an acceptable answer. Find the disagreement.

12. Vocabulary for this discussion

Term	Working definition in this context
Blast radius	How much live traffic a single command can affect before anyone notices.
Eventual consistency	A write is accepted before it is visible to queries. Verification needs a wait, not just a success code.
Idempotent	Running it twice leaves the same state as running it once. For an upsert, this depends entirely on deterministic record IDs.
Manifest	A durable record kept outside the index of what was indexed, from which source, at which version, with which model, under which IDs.
Orphan	A record in the index whose source document can no longer be located.
Soft delete	Mark a record inactive in metadata and filter it out at query time. Requires every consumer to apply the filter, every time.
Hard delete	Remove the record from the index. Requires knowing which records to remove.
Blue-green	Build the replacement alongside the original, evaluate both on the same queries, switch only if the new one wins, keep the old one for rollback.